Comparative Analysis of Image Compression: JPEG vs. JPEG2000

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Abstract

This report presents a comprehensive comparative analysis of two prominent image compression standards: JPEG and JPEG2000. We evaluate the performance of these compression algorithms across various standard datasets (Kodak, McM, SIPI) using quantitative metrics derived from information theory and signal processing, specifically Entropy, Redundancy, Mean Squared Error (MSE), and Energy Compaction. The goal is to determine the efficiency of the Discrete Cosine Transform (DCT) versus the Discrete Wavelet Transform (DWT) in representing image data with minimal information loss.

1 Problem Definition

The fundamental problem addressed in this study is the evaluation and comparison of JPEG and JPEG2000 image compression algorithms. As digital imaging continues to grow in resolution and volume, efficient storage and transmission require robust compression. While JPEG remains the most widely used standard, JPEG2000 was introduced to overcome its limitations, particularly regarding information preservation at high compression ratios. This report aims to quantitatively compare these two algorithms to determine their relative performance in terms of reconstruction error and information density.

2 Background and Theory

2.1 JPEG (Discrete Cosine Transform)

JPEG compression primarily works by removing information that the human eye cannot easily perceive. It processes an image in small, independent 8×8 pixel blocks:

- **Color Subsampling:** Color images are converted to brightness and color channels. Since our eyes are less sensitive to fine color details, the color channels are shrunk down, immediately saving space.
- **Discrete Cosine Transform (DCT):** Each 8×8 block is mathematically transformed from spatial pixels into 64 frequency "coefficients". The top-left value represents the block's average brightness, while the rest represent increasing levels of detail (high frequencies).
- **Quantization:** This is where actual compression (and quality loss) happens. The coefficients are divided by a step size Q and rounded. Because high-frequency details are less visible, they are divided by larger numbers, turning many of them into zeros.
- **Entropy Coding:** The resulting numbers are grouped using a zig-zag pattern (which collects all the zeros together). They are then shrunk losslessly using Run-Length Encoding (RLE) to count the zeros, and Huffman Coding, which assigns shorter binary codes to the most frequent numbers.

Limitation: Because JPEG processes images in strict 8×8 blocks, heavy compression causes the edges of these blocks to stop matching up, creating noticeable "blocking artifacts."

2.2 JPEG2000 (Discrete Wavelet Transform)

To fix JPEG's blocky artifacts, JPEG2000 processes the image as one large piece (or in very large tiles) using a different mathematical approach:

- **Discrete Wavelet Transform (DWT):** Instead of blocks, JPEG2000 passes the entire image through a series of filters that separate it into a blurred, shrunken version (LL) and sharp detail maps (edges and textures). This process is repeated multiple times on the blurred version to create a ”multi-resolution” image.
- **Quantization:** Similar to JPEG, the wavelet details are divided and rounded to discard invisible information. However, JPEG2000 allows for much finer control over how much detail is kept in different parts of the image.
- **Encoding:** The quantized details are flattened, run-length encoded, and compressed into a binary stream. Modern JPEG2000 uses complex bit-plane coding (EBCOT), though simpler Huffman coding can also be applied.

Advantage: By analyzing the whole image rather than rigid blocks, the DWT is much better at keeping sharp edges intact. At high compression levels, JPEG2000 degrades smoothly by getting slightly blurry, which looks much more natural to the human eye than JPEG's hard block edges.

3 Methodology

To systematically compare JPEG and JPEG2000, we established a robust evaluation methodology:

1. **Data Selection:** Standard test image datasets (Kodak, McM, SIPI) are utilized to capture a broad array of image statistics and spatial frequencies. Kodak provides 24 uncompressed true-color natural images widely established for evaluating compression artifacts; the McMaster (McM) dataset offers 18 high-resolution color images with distinct spectral characteristics compared to Kodak; and the USC-SIPI image database includes diverse classic images featuring rigorous textures, edges, and frequencies to thoroughly benchmark performance.

2. **Compression Execution:** Each algorithm is implemented in Python capturing their core standard pipelines. JPEG partitions the image into 8x8 blocks, applies 2D Discrete Cosine Transform (DCT), block quantization, zig-zag scanning, RLE, and Huffman coding. JPEG 2000 uses multi-level 2D Discrete Wavelet Transform (DWT) via the CDF 9/7 bior4.4 wavelet, dead-zone uniform scalar quantization, RLE subband flattening, and Huffman coding. In both algorithms, the quantization step size Q is varied to generate a spectrum of varying bitrates and compression levels.

3. **Metric Calculation:** We compute the following objective metrics:

- **Compression Ratio (CR):** The ratio of the uncompressed file size to the compressed file size.
- **Bits Per Pixel (bpp):** Measures the average number of bits required to represent a single pixel in the compressed image.
- **PSNR (Peak Signal-to-Noise Ratio):** Measures the pixel-level fidelity. Higher is better.
- **Mean Squared Error (MSE):** Measures the average squared difference between the original and reconstructed pixels.
- **Entropy (H):** Quantifies the average information content in the compressed image. Lower entropy at a given quality level indicates more efficient coding.
- **Redundancy (R):** Measures the fraction of data that is repetitive or predictable. It is defined as $R = 1 - \frac{1}{C}$, where C is the compression ratio.
- **Energy Compaction:** Evaluates the ability of the transform (DCT vs. DWT) to pack the majority of the image energy into a small number of coefficients.

4 Implementation Details

The compression pipelines were implemented from scratch in Python to faithfully replicate and analyze the theoretical mathematical underpinnings of each standard, rather than relying on black-box standard libraries.

- **JPEG Implementation:** The pipeline natively pads images to be divisible by 8 and partitions them into 8×8 blocks. We apply the 2D DCT on each block, perform uniform scalar quantization adjusted by a step parameter Q, and extract coefficients utilizing a custom zig-zag scan to group high-frequency zeros. The flat 1D array is then losslessly compressed using custom Run-Length Encoding (RLE) and a dynamically built Huffman entropy coder.

- **JPEG2000 Implementation:** The image undergoes a 3-level 2D DWT adopting the CDF 9/7 biorthogonal wavelet (`bior4.4`). The resulting coarse approximations (LL) and detail subbands are processed using Dead-Zone Uniform Scalar Quantization scaled by Q. The subbands are then flattened to group spatial zeros, run-length encoded, and transformed into a tight bitstream via dynamic Huffman coding.
- **Evaluation and Logging:** Automated routines iterate through the `kodak_dataset`, `McM_dataset`, and `sipi_dataset` directories, systematically modulating the quantization step size Q to sweep across varying bitrates. Custom functions calculate theoretical Bits Per Pixel (bpp) and Entropy from the simulated Huffman bitstream lengths, matched against reconstruction PSNR. Aggregated stats are saved out directly to CSV files (e.g., `Kodak_jpeg_avg_results.csv`) for plotting rate-distortion curves.

5 Experimental Setup

The experiments evaluating the compression schemes were performed on three renowned image datasets ensuring a wide variety of textural and structural features:

- **Kodak Dataset:** Consists of natural color images, widely used for visual quality evaluation.
- **McM (McMaster) Dataset:** Contains images with varying spatial frequencies and vibrant color contents.
- **SIPI Dataset:** Standardized test images (including classics like Lena, Mandrill) which provide a reference baseline for algorithm research.

5.1 Parameter Selection

To systematically benchmark the algorithms, we modulated the quantization step size parameter (Q). For the JPEG pipeline, Q values were linearly swept ranging from broad quantizations to negligible steps. For the JPEG2000 pipeline, compatible quantizations were structured so that both schemes spanned similar compression regimes.

5.2 Data Collection and Visualization

During the evaluation, specific objective metrics were automatically extracted for each image at every quantization level:

- **Primary Data Collected:** Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Compression Ratio (CR), theoretical Bits Per Pixel (BPP) computed from the simulated custom bitstream, and Entropy. Full raw statistics for each image run and dataset-wide averaged statistics were outputted locally to CSV logs (e.g., `Kodak_jpeg_full_results.csv` & `Kodak_jpeg_avg_results.csv`).
- **Generated Plots:** Extracted directly from the averaged CSV logs, four primary visualization charts were graphed uniquely for each dataset:
  1. **Rate-Distortion Performance (PSNR vs. BPP):** Plots the absolute reconstruction fidelity against the available bitrate.
  2. **Compression Quality (CR vs. PSNR):** Plots the geometric scaling of the reconstruction quality (PSNR) versus the mathematical compression ratio applied.
  3. **Entropy Analysis (CR vs. Entropy):** Demonstrates the reduction of information density as files are compressed tighter.
  4. **Energy Compaction Curves:** Plots cumulative transform energy retained versus coefficient fraction for DCT and DWT to compare transform coding efficiency.
- **Inferences Drawn:** We draw conclusions only from the generated figures: global overlays compare both algorithms across all three datasets, while dataset-level plots and energy curves provide transform-level evidence for quality and compression behavior.

6 Results and Analysis

Table 1 summarizes the typical average results achieved by JPEG and JPEG2000, as directly extracted from the CSV datasets. The table contrasts the resulting reconstruction error metrics alongside their file sizes collectively for all three evaluation sets.

Table 1: Average Performance Metrics across Kodak, McM, and SIPI Datasets

	JPEG (DCT)						JPEG2000 (DWT)					
Dataset	**Factor**	**PSNR**	**CR**	**BPP**	**MSE**	**Entropy**	**Q Level**	**PSNR**	**CR**	**BPP**	**MSE**	**Entropy**
Kodak	0.2	37.80	2.19	3.78	11.01	0.997	2	40.97	1.22	6.67	5.20	0.996
	0.5	34.97	3.74	2.24	22.41	0.997	5	39.73	2.08	4.06	6.91	0.996
	1.0	32.81	5.75	1.46	38.31	0.996	10	37.65	3.46	2.55	11.22	0.995
	2.0	30.71	8.94	0.94	63.16	0.995	20	34.73	5.90	1.55	22.54	0.994
McM	0.2	38.46	1.79	4.68	9.44	0.997	2	40.98	0.92	9.03	5.19	0.996
	0.5	36.13	3.08	2.80	16.99	0.997	5	39.70	1.84	4.77	6.99	0.996
	1.0	34.25	4.71	1.85	27.09	0.997	10	37.82	3.27	2.85	10.88	0.995
	2.0	32.17	7.22	1.20	44.04	0.996	20	35.18	5.76	1.65	20.55	0.994
SIPI	0.2	37.92	2.55	4.40	11.60	0.997	2	43.85	1.52	9.41	3.86	0.996
	0.5	34.97	4.20	2.50	24.42	0.996	5	40.99	2.29	5.62	6.15	0.994
	1.0	33.07	6.42	1.57	38.77	0.996	10	37.90	3.61	3.27	11.63	0.993
	2.0	31.26	10.30	0.95	58.80	0.995	20	34.57	6.46	1.76	24.83	0.992

6.1 Mean Squared Error (MSE) Comparison

The rate-distortion plots for each dataset (PSNR and MSE) show that JPEG2000 consistently preserves higher quality at equivalent bitrates. This is evident in Figs. 1, 2, and 3, where JPEG2000 achieves higher PSNR and lower MSE compared to JPEG across most compression levels, particularly in the low-bitrate regime.

6.2 Entropy and Redundancy

As expected, redundancy decreases with increasing compression ratio. The entropy-compression curves in Figs. 1, 2, and 3 illustrate how both methods reduce information content under stronger compression, while also highlighting that JPEG2000 maintains more efficient coding behavior compared to JPEG across all datasets.

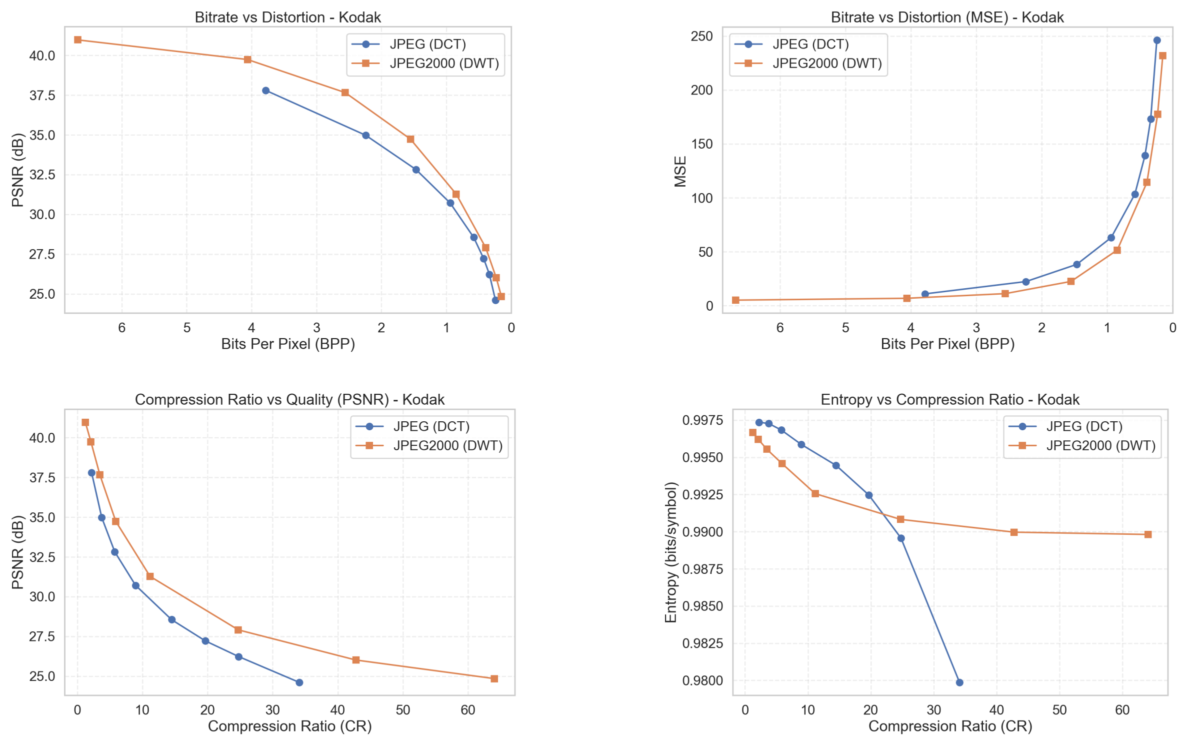

Figure 1: Performance comparison of JPEG and JPEG2000 on Kodak dataset.

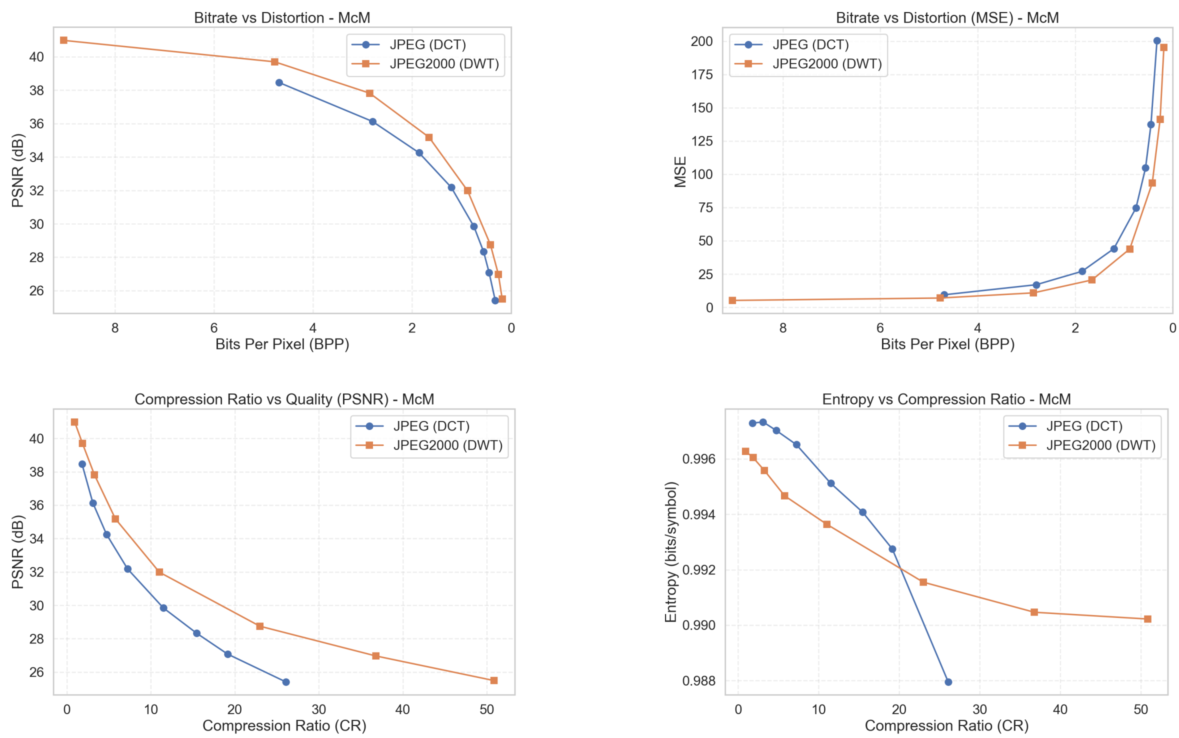

Figure 2: Performance comparison of JPEG and JPEG2000 on McMaster dataset.

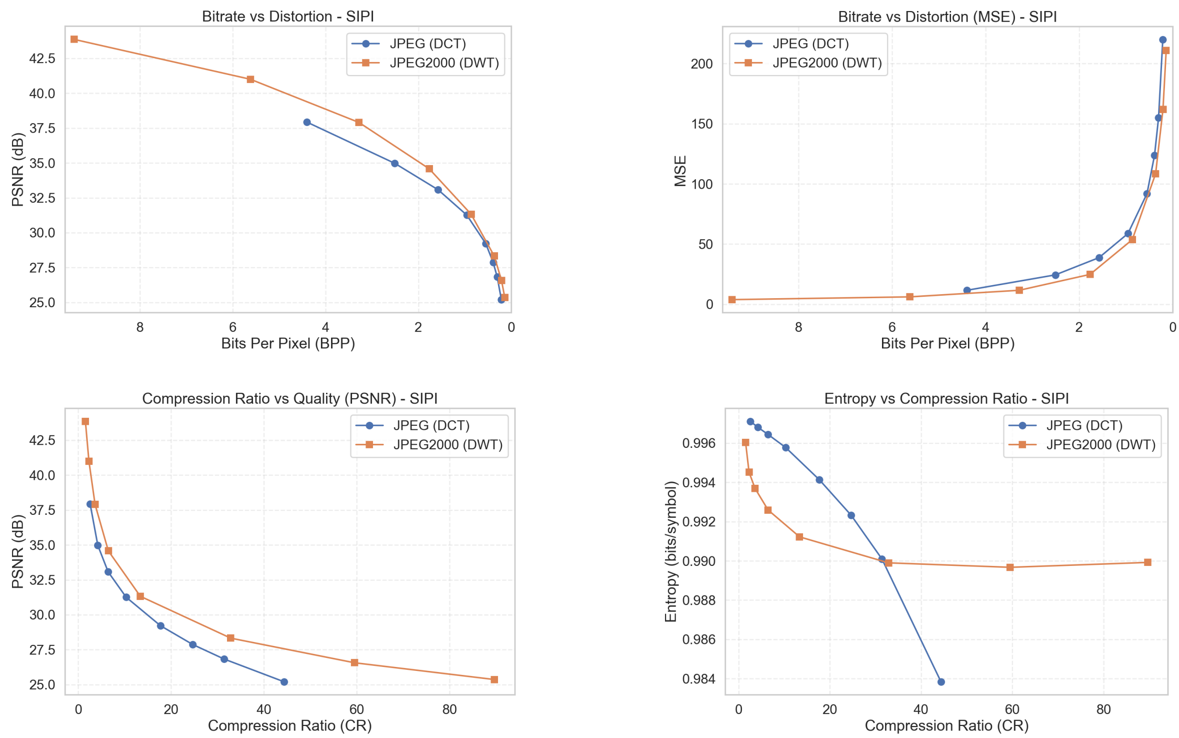

Figure 3: Performance comparison of JPEG and JPEG2000 on SIPI dataset.

6.3 Energy Compaction Efficiency

The energy compaction curves (Fig. ??) show that both JPEG (DCT) and JPEG2000 (DWT) concentrate most of the signal energy within a very small fraction of coefficients, with nearly 95% of the total energy captured within the first 2% of coefficients.

Interestingly, JPEG demonstrates slightly stronger energy compaction in the very early region, indicating that the DCT is highly effective at concentrating energy within a few low-frequency components in local blocks. However, JPEG2000 achieves comparable energy retention shortly after, reflecting its ability to distribute energy more uniformly across multiple resolution scales.

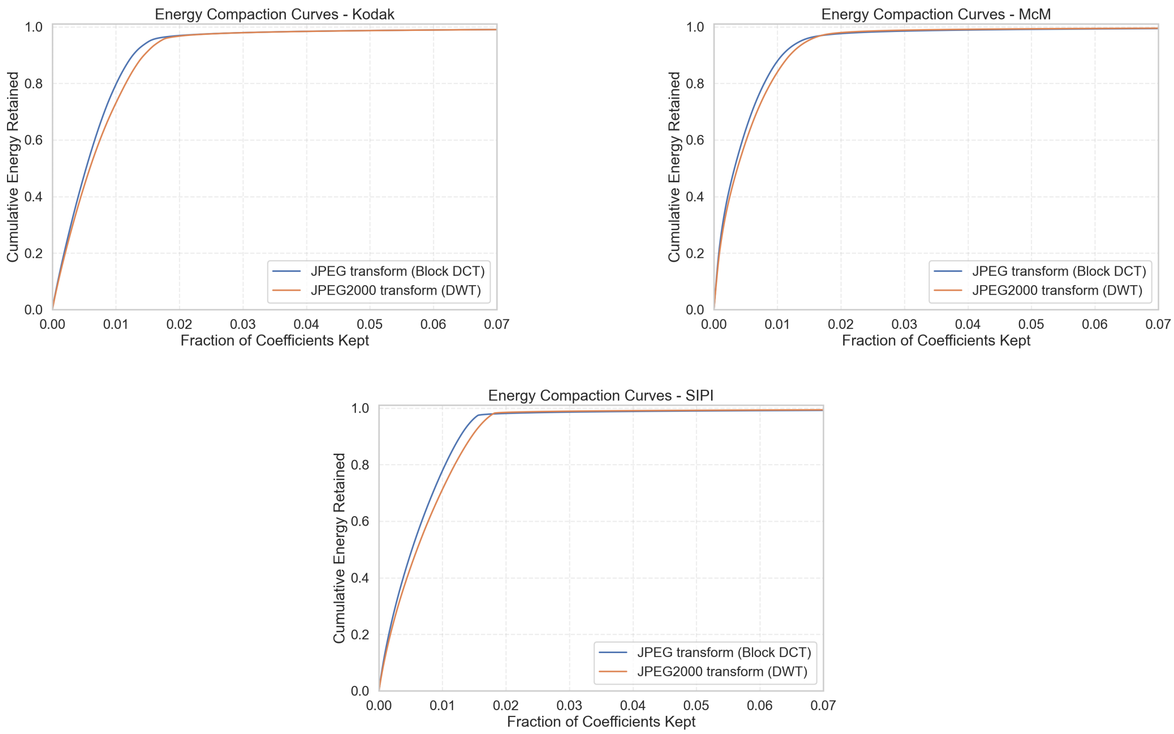

Figure 4: Energy compaction curves for transform coding on representative images from each dataset. Earlier-rising curves indicate stronger concentration of energy in fewer coefficients.

7 Key Insights

- **Error Minimization:** Across all datasets, the rate-distortion curves show JPEG2000 retains higher PSNR at matched compression levels.
- **Information Density:** Entropy-vs-compression plots provide direct evidence of information reduction trends for both algorithms over the full operating range.
- **Transform Efficiency:** The energy-compaction curves show DWT concentrates signal energy earlier than block-DCT, supporting improved quality at lower bitrate.
- **Computational Trade-off:** Despite the lower MSE and better compaction of JPEG2000, the algorithm requires higher computational overhead to calculate the multi-level DWT.

8 Conclusion

This comparative study highlights the efficiency of JPEG2000 in terms of information theory. By optimizing energy compaction and minimizing reconstruction error (MSE), JPEG2000 proves to be the superior standard for applications where minimizing information loss and maximizing data thinning (redundancy removal) are critical. While JPEG remains efficient for high-speed, general-purpose use, JPEG2000 provides a more compact and mathematically accurate representation of digital imagery.